THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3516	Apply the divisibility rule of 4.
2	**(d)** 8 and 15	Co-prime numbers have no common factor other than 1.
3	**(a)** 11 and 13	Twin primes are primes that differ by 2.
4	**(c)** 6	Check each number.
5	**(b)** 5	$72 = 2 \times 2 \times 2 \times 3 \times 3$
6	**(c)** 3	The next un-crossed number after 2 is 3.
7	**(a)** 3	Test 2, 3, 5, 7, ...
8	**(b)** 5128	Apply the divisibility rule of 8.
9	**(d)** 14 and 25	Co-prime numbers have no common factor other than 1.
10	**(a)** 17 and 19	Twin primes are primes that differ by 2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Last two digits 16 give divisible by 4 (1); Last three digits 216: divisible by 8 (1).	1+1
12	Try prime pairs (1); 5 + 19 = 24 (1).	1+1
13	Factors of 14: 1, 2, 7, 14; of 15: 1, 3, 5, 15 (1); Co-prime (only common factor is 1) (1).	1+1
14	Last two digits 24 give divisible by 4 (1); Last three digits 424: divisible by 8 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum of digits at odd places and even places (1); Difference $= 11$ (1); Divisible by 11 (1).	1+1+1
16	Split the number into two factors repeatedly (1); $96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$ (1); Product check: 96 (1).	1+1+1
17	LCM $= 12$ (1); Common multiples: 12, 24, 36, 48 (1); Each is a multiple of the LCM (1).	1+1+1
	OR: Sum of digits at odd places and even places (1); Difference $= 0$ (1); Divisible by 11 (1).	1+1+1
18	Split the number into two factors repeatedly (1); $108 = 2 \times 2 \times 3 \times 3 \times 3$ (1); Product check: 108 (1).	1+1+1
	OR: LCM $= 24$ (1); Common multiples: 24, 48, 72 (1); Each is a multiple of the LCM (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cross out multiples of 2, 3, 5, 7 (1); Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37 (1); Number of primes $= 12$ (1); Check by testing factors (2).	1+1+1+2
	OR: 3 and 4 have no common factor (1); Example: $60 \div 3 = 20$ and $\div 4 = 15$ (1); $60 \div 12 = 5$ (1); So both factors divide it and their product also divides (2).	1+1+1+2
20	Cross out multiples of 2, 3, 5, 7 (1); Primes: 41, 43, 47, 53, 59, 61, 67, 71, 73, 79 (1); Number of primes $= 10$ (1); Check by testing factors (2).	1+1+1+2
	OR: 4 and 5 have no common factor (1); Example: $140 \div 4 = 35$ and $\div 5 = 28$ (1); $140 \div 20 = 7$ (1); So both factors divide it and their product also divides (2).	1+1+1+2

Mistakes to avoid:

- Calling 1 a prime number; 1 is neither prime nor composite.
- Thinking all odd numbers are prime (9, 15, 21, 91 are composite) or that 2 is not prime.
- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 7	Check each number.
2	**(d)** 5	$120 = 2 \times 2 \times 2 \times 3 \times 5$
3	**(a)** 7	Primes p with $p \times p \leq n$.
4	**(d)** 11	Test 2, 3, 5, 7, ...
5	**(c)** 4218	Apply the divisibility rule of 6.
6	**(c)** 9 and 28	Co-prime numbers have no common factor other than 1.
7	**(b)** 29 and 31	Twin primes are primes that differ by 2.
8	**(a)** 7	Check each number.
9	**(d)** 6	$144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$
10	**(c)** 7	Primes p with $p \times p \leq n$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Try prime pairs (1); 5 + 31 = 36 (1).	1+1
12	Factors of 18: 1, 2, 3, 6, 9, 18; of 27: 1, 3, 9, 27 (1); Not co-prime (common factor 9) (1).	1+1
13	Last two digits 48 give divisible by 4 (1); Last three digits 348: not divisible by 8 (1).	1+1
14	Try prime pairs (1); 5 + 43 = 48 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum of digits at odd places and even places (1); Difference $= 10$ (1); Not divisible by 11 (1).	1+1+1
16	Split the number into two factors repeatedly (1); $126 = 2 \times 3 \times 3 \times 7$ (1); Product check: 126 (1).	1+1+1
17	LCM $= 10$ (1); Common multiples: 10, 20, 30, 40 (1); Each is a multiple of the LCM (1).	1+1+1
	OR: Sum of digits at odd places and even places (1); Difference $= 11$ (1); Divisible by 11 (1).	1+1+1
18	Split the number into two factors repeatedly (1); $150 = 2 \times 3 \times 5 \times 5$ (1); Product check: 150 (1).	1+1+1
	OR: LCM $= 36$ (1); Common multiples: 36, 72, 108 (1); Each is a multiple of the LCM (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cross out multiples of 2, 3, 5, 7 (1); Primes: 61, 67, 71, 73, 79, 83, 89, 97 (1); Number of primes $= 8$ (1); Check by testing factors (2).	1+1+1+2
	OR: 5 and 6 have no common factor (1); Example: $90 \div 5 = 18$ and $\div 6 = 15$ (1); $90 \div 30 = 3$ (1); So both factors divide it and their product also divides (2).	1+1+1+2
20	Cross out multiples of 2, 3, 5, 7 (1); Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (1); Number of primes $= 10$ (1); Check by testing factors (2).	1+1+1+2
	OR: 3 and 8 have no common factor (1); Example: $96 \div 3 = 32$ and $\div 8 = 12$ (1); $96 \div 24 = 4$ (1); So both factors divide it and their product also divides (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 13	Test 2, 3, 5, 7, ...
2	**(a)** 2728	Apply the divisibility rule of 11.
3	**(d)** 21 and 22	Co-prime numbers have no common factor other than 1.
4	**(b)** 41 and 43	Twin primes are primes that differ by 2.
5	**(c)** 8	Check each number.
6	**(a)** 4	$210 = 2 \times 3 \times 5 \times 7$
7	**(c)** 11	Primes p with $p \times p \leq n$.
8	**(c)** 17	Test 2, 3, 5, 7, ...
9	**(a)** 7924	Apply the divisibility rule of 4.
10	**(b)** 59 and 61	Twin primes are primes that differ by 2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Factors of 25: 1, 5, 25; of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36 (1); Co-prime (only common factor is 1) (1).	1+1
12	Last two digits 36 give divisible by 4 (1); Last three digits 136: divisible by 8 (1).	1+1
13	Try prime pairs (1); 3 + 53 = 56 (1).	1+1
14	Factors of 21: 1, 3, 7, 21; of 49: 1, 7, 49 (1); Not co-prime (common factor 7) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum of digits at odd places and even places (1); Difference $= 11$ (1); Divisible by 11 (1).	1+1+1
16	Split the number into two factors repeatedly (1); $198 = 2 \times 3 \times 3 \times 11$ (1); Product check: 198 (1).	1+1+1
17	LCM $= 21$ (1); Common multiples: 21, 42, 63 (1); Each is a multiple of the LCM (1).	1+1+1
	OR: Sum of digits at odd places and even places (1); Difference $= 0$ (1); Divisible by 11 (1).	1+1+1
18	Split the number into two factors repeatedly (1); $240 = 2 \times 2 \times 2 \times 2 \times 3 \times 5$ (1); Product check: 240 (1).	1+1+1
	OR: LCM $= 24$ (1); Common multiples: 24, 48, 72 (1); Each is a multiple of the LCM (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cross out multiples of 2, 3, 5, 7 (1); Primes: 31, 37, 41, 43, 47, 53, 59, 61, 67 (1); Number of primes $= 9$ (1); Check by testing factors (2).	1+1+1+2
	OR: 7 and 9 have no common factor (1); Example: $126 \div 7 = 18$ and $\div 9 = 14$ (1); $126 \div 63 = 2$ (1); So both factors divide it and their product also divides (2).	1+1+1+2
20	Cross out multiples of 2, 3, 5, 7 (1); Primes: 53, 59, 61, 67, 71, 73, 79, 83, 89 (1); Number of primes $= 9$ (1); Check by testing factors (2).	1+1+1+2
	OR: 5 and 8 have no common factor (1); Example: $240 \div 5 = 48$ and $\div 8 = 30$ (1); $240 \div 40 = 6$ (1); So both factors divide it and their product also divides (2).	1+1+1+2

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